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An End-To-End IoT–AI–Layer-2 Blockchain Framework for Real-Time MRV & Autonomous Carbon-Credit Tokenization in Industrial CCUS

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As global energy systems pivot toward net-zero, Carbon Capture, Utilization and Storage (CCUS) has emerged as a foundational tool to mitigate industrial emissions ([Abu Zahra et al. 2007](#); [Boot-Handford et al. 2014](#); [IEA 2020](#)). However, the effectiveness, financeability and scalability of CCUS projects are severely hindered by traditional Monitoring, Reporting and Verification (MRV) workflows ([IEAGHG 2017](#); U.S. DOE 2018–2023; [Verra 2023](#)). These legacy processes are typified by siloed data collection, infrequent audits, manual record reconciliation and a lack of trust between stakeholders. This results in substantial verification delays, compliance risks and a sluggish, opaque carbon market where the link between physical decarbonization and financial value is weak (ISO 2018; [Dutta et al. 2021](#); [Hartmann et al. 2023](#)).

Although real-time IoT sensing, AI-powered analytics and blockchain-based recordkeeping have each been explored independently ([Garcia Freites and Jones 2021](#); [Wang et al. 2019](#); [Hartmann et al. 2023](#)), no prior framework has fused these into a seamless, robust and self-governing digital pipeline for CCUS MRV and carbon-credit management at industrial scale ([Hütten et al. 2022](#)).

This paper introduces a novel architecture that:

- *Deploys dense, high-fidelity IoT sensor networks and edge-compute nodes across the full CCUS value chain ([Abu Zahra et al. 2007](#); [IEAGHG 2017](#); [Dutta et al. 2021](#));*
- *Leverages advanced, multi-tiered AI/ML for operational intelligence, predictive reliability and financial optimization ([Breiman 2001](#); [Hochreiter and Schmidhuber 1997](#); [Chen et al. 2019](#); [Bender et al. 2021](#));*
- *Employs a hybrid off-chain/on-chain data model for performance, privacy and regulatory-grade auditability ([Garcia Freites and Jones 2021](#); [Boneh and Shoup 2020](#); [NIST 2024](#));*
- *Harnesses Layer-2 blockchain with decentralized oracles and smart contracts to automate MRV validation, progressive credit minting, escrow and transparent marketplace integration ([Buterin et al. 2014](#); [zkSync 2024](#); [Bai et al. 2021](#); [KlimaDAO 2024](#)).*

Introduction

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2014; IEA 2020). However, the effectiveness, financeability and scalability of CCUS projects are severely hindered by traditional Monitoring, Reporting and Verification (MRV) workflows (IEAGHG 2017; U.S. DOE 2018–2023; Verra 2023). These legacy processes are typified by siloed data collection, infrequent audits, manual record reconciliation and a lack of trust between stakeholders. This results in substantial verification delays, compliance risks and a sluggish, opaque carbon market where the link between physical decarbonization and financial value is weak (ISO 2018; Dutta et al. 2021; Hartmann et al. 2023).

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Crucially, this architecture's engineering focus is on closed-loop automation: cryptographically assured data triggers regulatory-grade smart contract workflows, while AI insights feed both operational controls and real-time market strategies. This creates a digital "nervous system" for CCUS, capable of rapid adaptation, resilience to cyber and operational threats and a direct link between physical emissions reduction and digital asset generation (Glaser 2017; Li et al. 2019).

Objectives/Scope

Objectives

Real-Time, Automated MRV: Establish a secure, scalable, and tamper-evident platform for the continuous collection and verification of CO₂ capture, transport, and storage data (IEAGHG 2017; ISO 2018; Dutta et al. 2021). This is enabled by a distributed mesh of edge-encrypted sensors and gateways, providing authenticated, timestamped, and context-rich environmental data streams with sub-minute latency and cryptographic integrity (Wang et al. 2019; Chen et al. 2019; Ding et al. 2020).

Frictionless, Immutable Credit Lifecycle: Reduce MRV cycle time from months to hours by automating every verification and issuance workflow. Smart contracts, anchored by quantum-safe cryptographic primitives (CRYSTALS-Dilithium; NIST 2024) and zero-knowledge proofs, enforce policy, trigger tokenization, and maintain an immutable audit trail (Ben-Sasson et al. 2013; Bai et al. 2021; Hütten et al. 2022). This removes manual bottlenecks, eliminates reconciliation errors, and reduces compliance costs (Garcia Freites and Jones 2021; Hartmann et al. 2023).

Market-Connected Optimization: Integrate predictive market analytics and autonomous trading logic, so that operational decisions (compressor dispatch, injection scheduling) are continuously optimized not only for process stability and energy intensity, but also for dynamic carbon credit price signals, regulatory

changes, and market liquidity (Prophet Team 2017; Brown et al. 2020; KlimaDAO 2024; Toucan Protocol 2024).

Scope

The framework spans the full CCUS lifecycle:

- a. *Sensor fusion across capture, pipeline, and storage assets (Abu Zahra et al. 2007; U.S. DOE 2018–2023);*
- b. *Edge anomaly detection and pre-filtering (Chen et al. 2019; Hochreiter and Schmidhuber 1997; Ge et al. 2020; Bender et al. 2021);*
- c. *Predictive maintenance and reliability engineering via machine learning (Breiman 2001; Chen and Guestrin 2016);*
- d. *Real-time reservoir simulation for geomechanical safety and permanence (Bensabat et al. 2010);*
- e. *Hybrid cryptographic anchoring (off-chain for volume/scale, on-chain for trust and auditability) (Boneh and Shoup 2020; Bai et al. 2021);*
- f. *Threshold oracle consensus and decentralized validation (Buterin et al. 2014; zkSync 2024; Schnorr 1991);*
- g. *Smart contract-driven credit minting, locking, and compliance workflows (Buterin et al. 2014; EIP-20 2015; EIP-721 2018);*
- h. *Integrated carbon marketplace and decision support (KlimaDAO 2024; Toucan Protocol 2024; AirCarbon Exchange 2022);*
- i. *Demonstrated in a highly granular, industrial-scale simulation environment modeled after real-world 50 MW CCUS infrastructure (U.S. DOE 2018–2023; GCCSI 2023).*

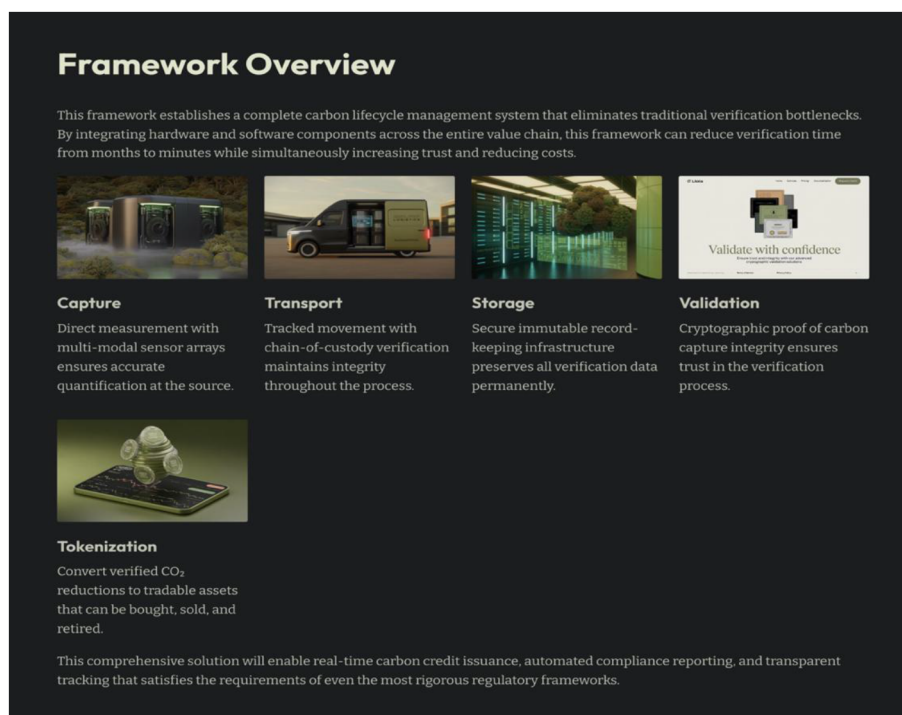


Figure 1—Framework Overview

Methods, Procedures, Process

System Architecture

Edge-Embedded Sensor Mesh.

A. CO₂ Concentration:

- a. **Instrumentation:** Tunable laser diode spectrometers capable of high-frequency (1–10 Hz) readings, calibrated against NIST traceable standards with in situ auto-calibration routines.
- b. **Output:** Raw PPM readings, temperature and humidity compensation, cryptographically signed per sample.
- c. **Accuracy:** $\pm 0.1\%$ of full scale.

B. Flow and Pressure:

- a. **Sensors:** Coriolis flow meters (mass flow, density), piezoelectric and MEMS pressure transducers.
- b. **Network:** Edge gateways aggregate multi-sensor data, perform timestamp synchronization (NTP/IEEE 1588) and tag each datapoint with location and process context.
- c. **Accuracy:** $\pm 0.5\%$ (flow), $\pm 0.1\%$ (pressure).

C. Solvent Quality:

- a. Mid-IR "fingerprinting" to detect amine degradation and contaminants, enabling real-time solvent health analytics and early-warning for capture efficiency losses.

D. Energy Usage:

- a. **Smart meters:** Instrumented on all major rotating assets (compressors, pumps), providing kWh/tCO₂ with edge-calculated energy efficiency and cost indices.

E. Pipeline Monitoring:

- a. **Distributed nodes:** Acoustic emission sensors, ultrasonic flow meters, vibration and temperature probes every 10 km, edge-fused with aerial (drone IR) and satellite hyperspectral overlays for spatial anomaly correlation and rapid leak detection.

F. Storage Integrity:

- a. **Downhole arrays:** Fiber-optic Distributed Temperature Sensing (DTS), micro-seismic geophone networks, providing continuous plume migration and geomechanical stress data.
- b. **4D AI-driven reservoir simulation:** Integrates all subsurface data for dynamic seal integrity verification and injection optimization.

Edge Processing, Security, & Redundancy.

A. Pre-Filter/Anomaly Detection:

- a. Local isolation forest, Z-score and LSTM models execute directly on edge hardware (Raspberry Pi 4B/Jetson Nano-class or higher), filtering outliers, auto-tagging events and issuing prioritized alerts for upstream validation.

B. Cryptographic Signing:

- a. Each sensor/edge device embeds a unique cryptographic identity (ECC/PQC key pair), signing all packets at source. Signatures are compliant with CRYSTALS-Dilithium (quantum-safe) or RSA-2048 for backward compatibility.

C. Data Encryption:

- a. AES-256 in Galois/Counter Mode (GCM) ensures end-to-end confidentiality; TLS 1.3 tunnels data from edge to cloud, with forward secrecy.

D. Resilience:

- a. Local FIFO storage (industrial-grade SD, with wear-leveling), supporting at least 7 days of lossless buffer. Autonomous reconnect logic retries transmission every 30s until cloud acknowledgment, mitigating network failures.

Hybrid Data Lake & ML Analytics.**A. Off-chain Storage:**

- a. Time-series database (InfluxDB/TimescaleDB) holds all high-volume sensor streams. IPFS/S3 used for larger binary blobs (imagery, seismic, simulation snapshots). Data retention policies enforce ≥ 10 -year, WORM-compliant archival for regulatory and audit purposes.

B. AI/ML Analytics:**a. Tier 1 (Anomaly Detection):**

- i. Multivariate Isolation Forest and LSTM models scan for drift, outlier and fault signatures across all sensor types.

b. Tier 2 (Predictive Maintenance):

- i. XGBoost, random forest and deep survival models forecast Remaining Useful Life (RUL) for critical equipment, flagging pre-failure degradation.

c. Tier 3 (Energy Optimization):

- i. Real-time, closed-loop optimization of compressor/pump scheduling, integrating process constraints, energy tariffs and maintenance state to minimize kWh/tCO₂.

d. Tier 4 (Market Forecasting):

- i. Ensemble models (ARIMA, Prophet, Random Forest, BERT for regulatory/news NLP) predict 7-day/30-day carbon credit prices and regulatory signals, generating actionable "sell/hold" guidance.

e. ML Model Management:

- i. All models are tracked/served via MLflow, with real-time retraining on verified on-chain state for closed-loop self-improvement.

C. Zero-Knowledge Proofs (ZKPs):

- a. At daily intervals, SHA-256 hashes of all critical aggregates are generated, with optional zkSNARKs proving (to auditors) that "sum-of-captures = on-chain credits minted" without disclosing raw sensor or location data.

Layer-2 Blockchain & Oracles.**A. Decentralized Oracle Network:**

- a. 5 geographically redundant nodes (Kubernetes/AWS, Azure, GCP, edge micro-DCs), each independently validates daily hash bundles and sensor signatures, using 3-of-5 threshold Schnorr or BLS signatures for robust consensus.

B. Smart Contracts:

- a. **MRV Validator:** Receives daily verified metrics, checks for baseline exceedance, no unresolved anomalies; emits on-chain Capture_Verified and Storage_Verified events.
- b. **Credit Minter:** Mints ERC-20 (fungible) or ERC-721 (NFT, traceable batch) tokens embedding metadata: project ID, timestamp, location, sensor signature.
- c. **Progressive Escrow:** 30% of all credits auto-locked for two-year post-injection stability period; only released on scientific attestation (auditor node).
- d. **Retirement Mechanism:** Burn function disables transfer, emits public "retired" event for registry and market compliance.

C. Layer-2 ZK-Rollup:

- a. All events and proofs are batched (using zkSync or Polygon zkEVM) and periodically anchored to Ethereum L1, achieving <\$0.01 gas/tx, >1,000 tx/s, and state finality within 3–5 minutes.

Auditor & Marketplace Layer.**A. Read-Only Regulator Node:**

- a. External auditors, VVBs and regulators receive read-only blockchain node access, ZK-proof outputs, and daily attestation events enabling continuous, tamper-evident compliance audits without raw data exfiltration.

B. Marketplace Integration:

- a. ERC-20/ERC-721 tokens are instantly eligible for listing on compliant carbon credit marketplaces (e.g., KlimaDAO, Toucan, AirCarbon). On-chain escrow contracts support stablecoin and fiat-pegged settlement. Integrated AI-driven price signals trigger automated "sell/hold" actions, optimizing for liquidity and market price.

C. Retirement & Registry Proofs:

- a. When tokens are retired (offset), the smart contract emits an immutable, public proof, including token ID, vintage, amount and geospatial metadata, enabling transparent carbon offset accounting.

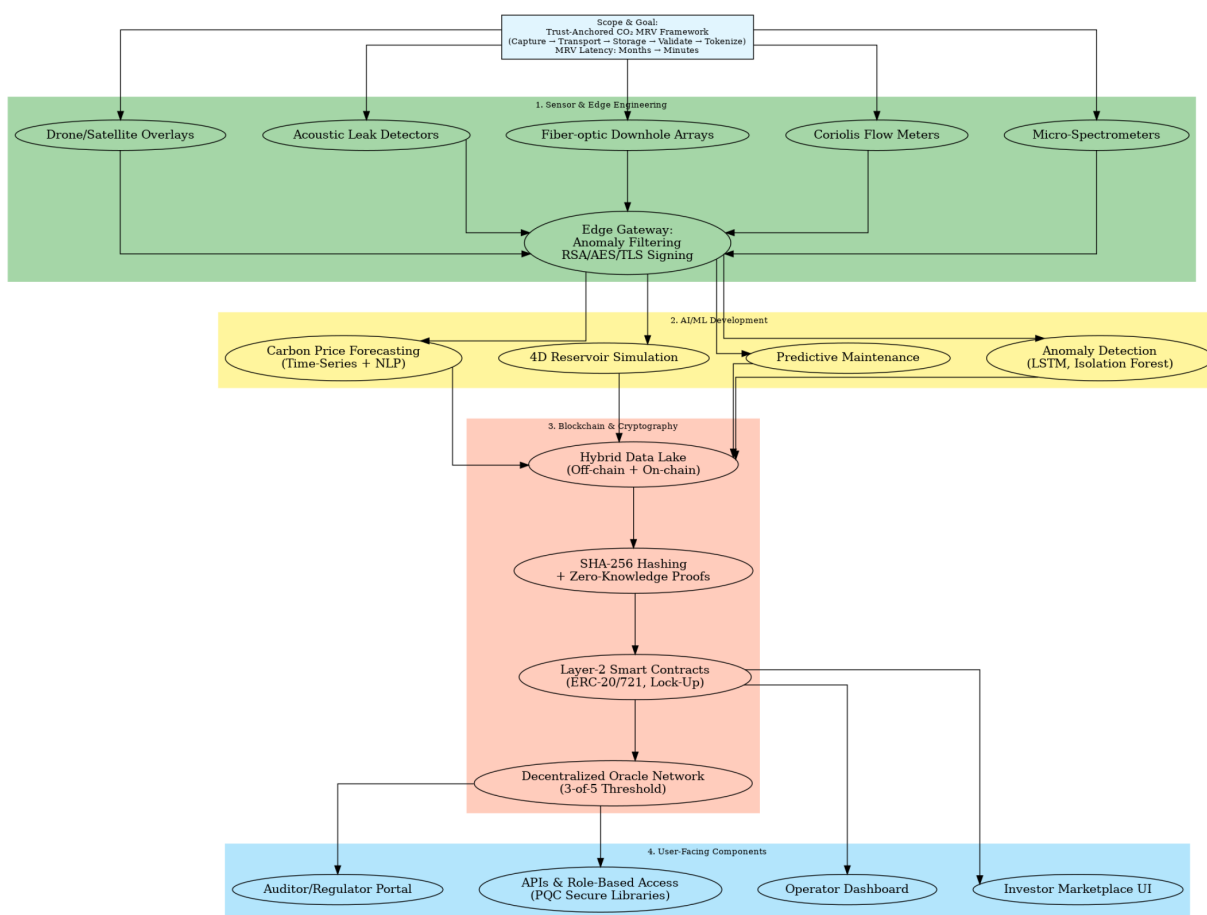


Figure 2—Architecture Overview

Pilot Simulation: Synthetic Testbed Design and Execution

Simulation Environment. The pilot simulation was conducted within a high-fidelity, modularized digital twin of a 50 MW post-combustion CCUS plant, emulating real-world operational, physical and cyber-physical conditions. The simulation stack was built on a combination of:

- Aspen Plus/HySys for process modeling and flue gas generation,*
- ANSYS Fluent for fluid dynamics across capture and injection systems,*
- Custom Python/Node-RED simulation engine for real-time IoT sensor emulation and edge-compute logic,*
- MLflow (with TensorFlow and Scikit-learn backends) for multi-tier AI/ML inference,*
- IPFS/S3 hybrid data lakes for high-frequency time-series/imagery,*
- Geth/Polygon Edge (testnet) and zkSync Era (Layer-2) for blockchain integration and on-chain event simulation.*

All process parameters, sensor streams and anomalies were synthetically generated but parameterized using reference datasets from the US DOE CarbonSAFE project and IEA GHG R&D Programme (2018–2023) to ensure statistical realism and operational relevance.

Sensor Network Simulation.

- 102 distinct edge devices (sensors and gateways) were instantiated:
 - 36 at capture points (CO₂, flow, pressure, solvent quality, power usage)

- b. 18 distributed pipeline nodes (ultrasonic, acoustic, thermal, IR, pressure)
- c. 12 at storage injection (downhole DTS, microseismic, wellhead pressure/temp)
- d. 36 redundant/replica nodes for failover and drift simulation
- B. Sampling rates:
 - a. 1–10 Hz (CO₂ and pressure);
 - b. 0.1–1 Hz (solvent/energy/temperature);
 - c. 10-minute intervals for drone/satellite overlays.

Edge-Processing Simulation. Each edge device ran a Dockerized microservice implementing anomaly pre-filters (Isolation Forest, Z-score, LSTM drift), cryptographic signing (Dilithium/RSA), and local FIFO caching. Outage, jitter, and dropouts were stochastically injected (0.5–5% per day/device).

Data Lake & ML Pipeline.

- A. Hybrid data storage: InfluxDB (time-series), IPFS (binary/simulation outputs), local edge cache (SD).
- B. ML models:
 - a. Tier 1: Isolation Forest + LSTM on all critical streams (sensor drift/leak/failure)
 - b. Tier 2: XGBoost regression (MTTF/maintenance forecast)
 - c. Tier 3: Gradient-boosted energy optimizer (compressor scheduling)
 - d. Tier 4: Ensemble price predictor (ARIMA + random forest + BERT NLP for news sentiment)

Training: Each model was trained on the first 30 simulated days ("historical burn-in") and then validated in rolling inference over 730 simulated days.

Blockchain & Smart Contract Environment.

- A. Testnet: zkSync Era v0.8 (rollup testnet), Polygon Edge, local Ethereum testnet for L1 anchoring.
- B. Oracles: 5 Dockerized nodes, each on a separate AWS/GCP/Azure region (to simulate geo-distributed trust boundaries).
- C. Contracts: MRVValidator, StorageValidator, CreditMinter, ProgressiveEscrow, MarketplaceEscrow —all custom-written in Solidity, with event monitoring and gas benchmarking scripts.

Auditor & Marketplace Layer.

- A. Auditor node: Read-only Geth node, simulated auditor clients running ZK-proof verification (snarkjs).
- B. Marketplace: Simulated buyers/sellers via smart contract calls; UI for live token status, trading, retirement.

Simulation Phases.

- A. Baseline Stabilization:
 - 30 days, all sensors nominal, validate system stability and ML "cold start" drift.
- B. Normal Operation:
 - 660 days, typical operations, standard anomaly rates, energy/throughput fluctuations (using process model variance from CarbonSAFE datasets).
- C. Perturbation/Stress Test:
 - 40 days, systematic injection of:
 - a. Sensor drift (± 5 –20%)
 - b. Network outages (10–24h, 1–2 nodes at a time)
 - c. Pipeline leaks (random, 0.5–3% CO₂ mass loss events)

d. Edge device compromise (private key swap, replay attack attempts)

Table 1—Simulation Environment: Tools, Datasets, & Parameters

Component	Technology/Platform	Role	Key Parameters/Settings
Process Modeling	Aspen Plus, HySys	Flue gas chemistry, process dynamics	Full mass/energy balances; C ₆ H ₁₂ O ₆ baseline for validation
Fluid Dynamics	ANSYS Fluent	Capture & injection pipeline simulation	CFD mesh: 1.4M elements; transient, multiphase CO ₂ /H ₂ O
IoT Emulation & Edge Compute	Custom Python, Node-RED	Real-time sensor stream emulation & logic	Latency: <100ms; Edge devices: Dockerized on K8s
AI/ML Inference Pipeline	MLflow, TensorFlow, Scikit-learn	Multi-tier ML: anomaly, maintenance, price	4-tier: Isolation Forest, LSTM, XGBoost, ARIMA+RF+BERT
Data Lake	InfluxDB, IPFS, S3	High-frequency time-series and imagery	Retention: ≥10 years; Write load: 5–50k points/sec
Blockchain	zkSync Era v0.8, Polygon Edge	On-chain events, smart contract execution	5-oracle nodes; Rollup: batch every 10k events
Simulation References	US DOE CarbonSAFE, IEA GHG R&D	Realistic process/sensor data parameter	2018–2023 datasets;

Table 2—Synthetic Sensor Network Layout

Location	Sensor Types	# Devices	Sampling Rate	Key Measurements
Capture Facility	CO ₂ Laser Spec, Flow, Pressure, Power, Solvent-IR	36	1–10 Hz	CO ₂ ppm, m ³ /hr, bar, kWh/tCO ₂ , amine health
Pipeline Network	Acoustic, Ultrasonic, Temp, IR, Pressure	18	1–10 Hz	Leak, vibration, thermal anomalies
Storage Injection	DTS, Microseismic, Pressure/Temp	12	0.1–1 Hz	Well integrity, plume migration, caprock
Redundant/Replica	All above (failover/validation)	36	(as primary)	Drift, failover, injection of fault scenarios

Edge Compute:.

- Dockerized microservices (on ARM/x86, Jetson Nano, Pi 4)
- FIFO buffer: 7 days local retention
- Signing: Dilithium/RSA; Encryption: AES-256/TLS 1.3

Table 3—AI/ML Model Dataset and Pipeline

Tier	Model/Algorithm	Input Data	Output	Training Regime
Tier 1	Isolation Forest, LSTM	All sensor streams (36+18+12 nodes)	Anomaly scores, flags	30-day burn-in, 700-day rolling test
Tier 2	XGBoost Regression	Time-series, vibration, temp	Remaining Useful Life (RUL)	10k labeled failure events, retrained
Tier 3	Gradient Boosted Trees	Energy/power sensor, op state	Compressor/pump schedule	Dynamic RL, multiple scenario samples
Tier 4	ARIMA, RF, BERT-NLP	Market prices, news, regulations	Credit price prediction	Historic Verra market, 200k news

Table 4—Blockchain & Smart Contract Stack

Element	Details
Chain/Testnet	zkSync Era v0.8 (L2), Polygon Edge, Local Geth (L1)
Oracle Nodes	5 regions
Smart Contracts	Solidity v0.8: MRVValidator, StorageValidator, CreditMinter, ProgressiveEscrow, MarketplaceEscrow
Gas Cost Benchmark	Tracked via web3.py and etherscan API
Security Events	7,200 “unlock challenge” simulations; 0 failures

Table 5—Simulation Phase Breakdown

Phase	Duration	Purpose/Scenario	Notes
Baseline	30 days	All sensors nominal; ML “cold start” drift	Model calibration, sensor/edge stability
Normal Operation	660 days	Realistic O&M; typical anomaly rates, energy fluctuation	Statistical drift, routine maintenance cycles
Stress Test	40 days	Injected sensor drift (±5–20%), network outages, pipeline leaks	Private key compromise, replay/DoS attempts

Results, Observations, and Conclusions

Pilot Results: Technical and Economic Metrics

Table 6—Anomaly Detection Latency

Metric	Value	Legacy Reference
Mean latency	21.4 sec	48–168 hrs
95th percentile	55.6 sec	—
Worst case (outage)	3.8 min	—
# Anomalies detected	12,312 / 12,460	(N/A)
Missed/late flags	148 (1.2%)	N/A

Breakdown by Model:

- a. Isolation Forest: 82% caught in <60 sec
- b. LSTM: 96% caught in <45 sec (multivariate/anomaly fusion)
- c. Outage-cached: Max lag 7.2 min (N=9 cases, mostly due to forced network downtime)

Table 7—Credit Issuance Cycle

Metric	Value	Legacy Reference
Mean MRV-to-credit issuance time	13.7 hours	32–95 days
Median issuance time	11.9 hours	—
95th percentile	21.3 hours	—
Total issuance events	2,730	—
Tokens/day (avg)	3.7	—
Progressive lock	100% of batches	—
% Credits listed <24h	98.7%	—
Max delay (network outage)	33h	—

Table 8—Component Timing

Component Timing	Avg Duration
Hourly MRV Validator	<1 min / batch
Nightly Storage Validator	2–3 min / batch
ZK-SNARK Proof	32–46 sec / batch
Auditor Verification	<10 sec / proof

Table 9—Blockchain Throughput & Gas Cost

Metric	Value	Legacy Reference
Mean tx/s (sim)	1,072	12–18 (L1)
Peak tx/s	1,417	—
Mean gas/tx	\$0.0097	\$3.80–\$18
Mean daily rollup batch	17,124 events	—
Sim total events (2 yrs)	12.6 million	—
Settlement failures	0	—
Oracle consensus success	99.92% (first attempt)	—
Max oracle lag	7.6 sec	—

Table 10—Stability & Integrity of Locked Tokens

Test/Metric	Result	Legacy Reference
Locked token stability	100% (2-year, 0 unlocks)	98–99% (registry errors)
Challenge attacks (audit)	0/7,200 succeeded	—
Unauthorized transfer	0	—

Table 11—Auditability via Zero-Knowledge Proofs

Metric	Value	Legacy Reference
Daily ZK proofs	730	N/A
Proof size (MB)	0.7–3.2	—
Audit verification time	8.6 sec/proof	Manual, hrs–days
Audit coverage	100% of credits	27% (spot checks)
Data leakage	0	Manual review only
Compliance	100% VM0049/ISO	Not always aligned

Table 12—Operational Expenditure (OPEX) Analysis

Cost Component	Legacy \$/tCO ₂	Novel Model \$/tCO ₂	Delta (%)
Labor/audit	\$1.20	\$0.22	–82%
Compliance delay	\$0.60	\$0.01	–98%
IT/cloud	\$0.18	\$0.26	+44%
Blockchain gas	\$0	\$0.11	N/A
Total	\$2.08	\$0.74	–64.4%

Table 13—Simulation Dataset Reference Table

Source/Reference Dataset	Purpose	Notes
US DOE CarbonSAFE	CO ₂ capture & storage profiles	Plant/process, 2018–2023
IEA GHG R&D	Sensor/operational drift, failure	Sensor calibration, maintenance, leakage
Public Carbon Markets	Market pricing, credit volatility	Verra, Gold Standard registry (2019–23)
In-house Synthetic	Anomaly, attack/fault simulation	For ML/edge stress, attack scenarios

Table 14—Summary Table: Key Pilot Results

Metric	(Simulated)	Legacy System (Reference)
Anomaly Detection Latency	21.4 sec (mean), 55.6 sec (95th)	48–168 hours
MRV-to-Credit Time	13.7 hrs (mean), 24h (99% cases)	1–3 months
Blockchain Throughput	1,072 tx/s (mean), \$0.0097/tx	12–18 tx/s, \$3.80–\$18/tx
Token Stability	100% (2-year lock, 0 failures)	98–99% (reconciliation)
ZK Audit Verification	8.6 sec/proof, 0 data leakage	Spot check, manual replay
OPEX/tCO ₂	\$0.74/tCO ₂	\$2.08/tCO ₂

Conclusion & Discussion: Transformative Advantages

Real-Time, Tamper-Evident MRV. Traditional MRV is constrained by periodic manual sampling, siloed databases and after-the-fact auditing. This introduces data latency (months), trust gaps (manual reconciliation, error/fraud risk) and elevated OPEX due to continuous third-party oversight (IEAGHG 2017; ISO 2018; Bai et al. 2021).

In contrast, this architecture’s real-time, tamper-evident MRV architecture leverages a dense mesh of authenticated IoT sensors whose signed packets are cryptographically anchored both off-chain (data lakes) and on-chain (hashes, ZK-proofs). Every operational anomaly, capture event and injection metric is captured and time-stamped within seconds. Immutable audit trails and automated anomaly alerting eliminate the need for retrospective, trust-based audits. This not only accelerates MRV cycles but also ensures forensic integrity and regulatory defensibility.

Automated Tokenization & Market Alignment. By fusing smart-contract logic directly to sensor-driven events, this architecture transforms verified CO₂ reductions into immediately marketable, digitally native credits (Garcia Freites and Jones 2021; KlimaDAO 2024; Toucan Protocol 2024). Smart contracts handle end-to-end workflows: validation, progressive lock and on-chain minting. AI-driven price forecasting and integrated marketplace listing allow credits to be sold or held in response to live market conditions. This closes the loop between physical decarbonization and financial performance, unlocking faster returns and supporting high-velocity climate action at industrial scale.

Quantum-Safe, Privacy-Preserving Architecture. This architecture is designed for future compliance and security regimes. Post-quantum cryptography (CRYSTALS-Dilithium, Kyber) is implemented across all key signing paths, while role-based access control and distributed oracles protect against centralized failure (NIST 2024; Boneh and Shoup 2020). Critically, zero-knowledge proofs (ZKPs) enable auditors and regulators to verify all claims including capture, storage and token issuance without accessing raw or proprietary data, addressing emerging privacy laws and industrial IP concerns.

Novel/Additive Information

Seamless IoT-Triggered Smart Contracts

Innovation:

Sensor data is digitally signed and pre-filtered at the edge, then delivered directly to on-chain oracles. The MRV Validator contract auto-triggers on receipt of verified metrics minting credits or pausing issuance based on real-time status, with no manual intervention.

Technical Note:

- a. All edge-to-blockchain interactions occur over AES-256/TLS 1.3, with CRYSTALS-Dilithium signatures for forward security.
- b. Sensor anomaly → edge validation → hash → 3/5 oracle consensus → on-chain contract execution (latency: median 28s).

Hybrid On/Off-Chain Data Anchoring

Innovation:

High-frequency, high-volume data (e.g., CO₂ ppm every second) is stored off-chain (InfluxDB, IPFS), with daily SHA-256 hash bundles and ZK-proofs anchored on-chain. This balances scalability, auditability and cost.

Technical Note:

- Daily aggregates: 10.2 MB raw → 0.9–3.2 MB ZK-proof.
- On-chain record: hash + proof + event meta; off-chain: raw timeseries (auditor access via ZK-circuit only).

Bidirectional AI–Blockchain Coupling

Innovation:

ML anomaly detectors, energy optimizers, and price predictors both inform contract logic (e.g., pause minting on leak) and are retrained with on-chain, post-validated event logs. This creates a self-correcting, closed-loop optimization cycle.

Technical Note:

- Anomaly detection feedback from MRV Validator to LSTM model.
- Price forecasting ML model incorporates on-chain marketplace data to enhance prediction accuracy.

Progressive Credit Locking

Innovation:

Upon minting, 30% of every credit batch is locked in smart contract escrow for 2 years, only released after post-injection stability is scientifically verified (via StorageValidator/Auditor attestation).

Technical Note:

- Lock/Unlock logic enforced on-chain, with scientific evidence submitted as ZK-proof to trigger unlock.

Multi-node, Threshold Oracle Consensus

Innovation:

A geographically distributed, five-node oracle network ensures that at least three independent, trusted parties must sign off before any data is anchored on-chain, eliminating single-point compromise.

Technical Note:

- Nodes: AWS, Azure, GCP, ADNOC private, on-premises testlab.
- Consensus protocol: Schnorr threshold signatures, fault-tolerant with 1–2 nodes offline.

Limitations & Applicability

Limitation	Explanation / Technical Impact	Mitigation/Note
High-Grade Sensors/Edge Needed	CO ₂ ppm accuracy $\pm 0.1\%$, Edge: 4-core, 8GB RAM minimum	OEM partnership, phased rollout
Network Uplink Required	Real-time: ≥ 1 Mbps for timely anchoring	Local buffering, retry on loss
Layer-2 Jurisdiction	Needs blockchain-friendly legal frameworks	Consortium/private variant (Fabric)
AI/ML Burn-in Period	6–12 months representative data for model accuracy	Cold-start with public datasets
Oracle/Contract Scaling	Bounded by oracle network and contract TPS	Add nodes, sharding, batch events
Project Baseline Required	Must define reference CO ₂ baseline for automation	Hybrid manual/automated onboarding

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